MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, XM267

1. SCOPE

1.1 This specification covers Cartridge, Impulse, XM267, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

D10551458	- Cartridge, Impulse, XM267
C9268094	- Box, Packing, Ammunition for Cartridge, Impulse, XM267
C9268092	- Box, Fiberboard, (Inner) Packing, Ammunition, for Cartridge, Impulse, XM267

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

DRAWINGS (Cont'd)

IEL 10551458 - Inspection Equipment List for Cartridge, Impulse, XM267

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply:

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts, and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The assembled cartridge shall withstand the vibration prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant, or derangement of any component.

3.4 Functional requirements.

3.4.1 The cartridge shall not fail to fire.

3.4.2 The cartridge when assembled and tested in an acceptable XM7 Parachute Ejector, temperature conditioned at -65° to -70° Fahrenheit (F), and at 160° to 165° F, shall be capable of moving a 10.75 pound weight along a horizontal path of travel and shall complete its stroke of 15 7/16 inches, minimum. The velocity of the mass upon reaching the end of the stroke shall be 57 feet per second (fps) minimum. The thrust shall not exceed 1700 pounds, maximum.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled cartridges for ballistic tests, 5 assembled cartridges for sealant application tests and 10 sets of unassembled components (including explosive components) for visual examination, and dimensional examinations and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specification at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic test shall be temperature conditioned and tested to determine compliance with 3.4.2. The sample for firing shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

TABLE I

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	3	4
65°F to 75°F	3	4
160°F to 165°F	3	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot
c. Primers from not more than one lot
d. Igniters from not more than one lot

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Table II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

TABLE II

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Primer, Percussion, Drawing D8798312 covering a detail of Drawing D10551458.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Ring, Retaining, Drawing C10551419 covering a detail of Drawing D10551458.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of center hole with outside diameter	Gage
102. Diameter, O.D.	Gage
103. Diameter of center hole, max.	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
MINOR	
201. Radius and angles	Gage
202. Diameter of center hole, min.	Gage
203. Thickness	Gage
204. Surface finish	Visual

4.3.2.3 Case, Cartridge, Drawing C11730413 covering a detail of Drawing D10551458.

CRITICAL	
1. Metal defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of ID counterbore with OD body	Gage
102. Concentricity of flash hole with primer pocket	Gage
103. Concentricity of primer pocket with OD body	Gage
104. Width of primer pocket	Gage
105. Depth of primer pocket	Gage
106. Thickness of web	Gage
107. Body, OD diameter	Gage
108. Diameter of counterbore in ID	Gage
109. Protective finish inadequate or defective	Visual
110. Burr or sharp edge	Visual
111. Foreign matter	
MINOR	
201. Total length	Gage
202. Depth of counterbore	Gage
203. Diameter of large inside cavity	Gage
204. Diameter of flash hole	Gage
205. Diameter of base	Gage
206. Thickness of base	Gage
207. Thickness, from base to bottom of ID cavity	Gage
208. Min. length to Rockwell	Gage
209. Radius or chamfer missing or incorrect	Gage/Visual
210. Surface finish improper	Visual

4.3.2.4 Disc, Sealing, Drawing B10551459 covering a detail of Drawing D10551458.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter	Gage
102. Thickness	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.4 (Cont'd)

Defect	Method of Inspection
103. Surface finish inadequate	Visual
104. Protective finish, missing or inadequate	Visual
105. Burr	Visual
106. Foreign matter	Visual

4.3.2.5 Cartridge, Impulse, XM267 Assembly Drawing D10551458.

CRITICAL

1. Metal defective	Visual
2. Primer anvil legs not seated to bottom of primer pocket	X-ray 1/ Visual
3. Improper assembly	X-ray 1/ Visual
4. Any part missing	X-ray 1/ Visual
5. Primer cup not flush or below flush with Surface "S"	Visual
6. Marking missing, incorrect or illegible	Visual

MAJOR

101. Crimp missing, improper or defective	Gage
102. Sealant missing or improperly applied	Visual

MINOR

201. Diameter of body, max.	Gage
202. Diameter of base, max.	Gage
203. Total length, max.	Gage
204. Moisture and fungus resistant varnish missing or inadequate	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.6 Carton, Packing, Ammunition For Cartridge Impulse XM267 Drawing C9268092.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

	Defect	Method of Inspection
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual/Visual
202.	Quantity of contents incorrect	Visual
203.	Separation or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damaged through all impregnated layers	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.7 Box, Packing, Ammunition For Cartridge Impulse XM267 Drawing C928094.

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual/Visual
202.	Box damaged	Visual
203.	Contents of box exposed	Visual
204.	Hardware missing, loose, broken or improperly engaged	Visual
205.	Strapping missing, loose broken or improperly assembled	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartons incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provision set forth in MIL-STD-105.

4.3.3.2 Vibration.-Twelve of the 21 cartridges of the initial production sample shall be subjected to vibration. The twelve cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.3 Hardness.-Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Airtightness.-Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.5 Sealant application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.-The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.-Twenty-one assembled cartridges from each production lot shall be functionally tested, as prescribed in Table IV. Failure of any cartridge to comply with the requirements of 3.4.2 shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample Size
-65°F to -70°F	7
65°F to 75°F	7
160°F to 165°F	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-10551458.

4.4 Test method and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawings.

4.4.4 Airtightness.-The method for determining airtightness shall be by use of the equipment listed on IEL-10551458.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant is free of sealant.

4.4.6 Ballistics.-The cartridge shall be loaded into an acceptable XM7 parachute ejector, divided into 3 equal groups and temperature conditioned in accordance with Table IV. The cartridge shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the thruster shall be reconditioned in accordance with 4.4.6.1.

4.4.6.1 Temperature conditioning.-A standard XM7 parachute ejector loaded with a dummy XM267 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C9268094 and C9268092.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

6.2 Storage and Shipping Information
Explosive Hazard Classification - Class 3
Storage Compatibility - Group B
Department of Transportation (DOT) Hazard Class; Class C
DOT Marking - "Explosive Power Device - Class C - Handle Carefully, Keep Fire Away".

Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1377-A502

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.